



NEXT CHAPTER @ SMITH PUBLIC LIBRARY

Numbers You Already Know

Session 2 — Math Part A: Whole Numbers, Fractions, Decimals & Percentages

Monday Evenings • 5:30–7:30 PM • Library Conference Room

Tonight's Agenda

1

Welcome & why math anxiety is normal

10 min

2

Order of operations (no calculator): concept + guided practice

30 min

3

Fractions, decimals, percentages: same number, three costumes

20 min

4

Conversion practice: guided problems

30 min

5

Worked example together

15 min

6

Wrap-up & take-home preview

15 min

Let's Name It: Math Anxiety Is Real



You already use math every day.

Making change, doubling a recipe, figuring out a discount — that's all math you already do.



We're not starting from zero.

Tonight builds on real number sense you already have, just with new labels.



Stuck is normal, not shameful.

If something doesn't click tonight, that's exactly what Thursday office hours are for.



NO CALCULATOR

Order of Operations

PEMDAS: the order matters.

Parentheses → Exponents → Multiply/Divide → Add/Subtract

Example: $12 + 8 \times 3$

Multiply first: $8 \times 3 = 24$

Then add: $12 + 24 = 36$

Why no calculator? The real GED opens with about 5 questions where no calculator is allowed — pure mental math. Tonight we build that muscle on purpose.

Your Turn: Try These

NO CALCULATOR

1

$$8 + 4 \times 2 = ?$$

2

$$(6 + 2) \times 3 = ?$$

3

$$20 - 6 \div 2 = ?$$

4

$$5 + 3^2 = ?$$

Let's Check Your Answers



1. $8 + 4 \times 2 = 8 + 8 = 16$



2. $(6 + 2) \times 3 = 8 \times 3 = 24$



3. $20 - 6 \div 2 = 20 - 3 = 17$



4. $5 + 3^2 = 5 + 9 = 14$

Same Number, Three Costumes

$\frac{1}{2}$

Fraction

0.5

Decimal

50%

Percentage

Converting Between Them: All 4 Directions

Fraction → Decimal

Divide top by bottom.

$$3/4 \rightarrow 3 \div 4 = 0.75$$

Decimal → Percentage

Move decimal 2 places right.

$$0.75 \rightarrow 75\%$$

Percentage → Decimal

Move decimal 2 places left.

$$75\% \rightarrow 0.75$$

Decimal → Fraction

Use the place value as denominator, then simplify.

$$0.75 = 75/100 = 3/4$$

Fraction ↔ Percentage (two steps): convert to a decimal first, then to a percentage. $1/8 \rightarrow 0.125 \rightarrow 12.5\%$

Your Turn: Convert These

GUIDED PRACTICE

1

Write $\frac{3}{5}$ as a decimal.

2

Write 0.6 as a percentage.

3

Write 80% as a decimal.

4

Write 0.125 as a fraction.

5

Write $\frac{1}{8}$ as a percentage.

Let's Check Your Answers



1. $\frac{3}{5} = 3 \div 5 = 0.6$



2. $0.6 \rightarrow$ move decimal right $\rightarrow 60\%$



3. $80\% \rightarrow$ move decimal left $\rightarrow 0.8$



4. $0.125 = \frac{125}{1000} = \frac{1}{8}$



5. $\frac{1}{8} = 0.125 \rightarrow 12.5\%$

Let's Solve One Together

A shirt is marked 25% off. The original price is \$40. What's the sale price?

Step 1: Find 25% of \$40 $\rightarrow 0.25 \times 40 = \10

Step 2: Subtract the discount $\rightarrow \$40 - \$10 = \$30$

Sale price: \$30

Before You Go



Try tonight's take-home practice

Real practice questions on tonight's material — 10 minutes, any time this week.



Thursday office hours are open

If conversions didn't click tonight, this is the place to ask.



Next week: more Math

Ratios and percentage applications — building on tonight.



You Just Did Real Math.

See you next Monday — ratios and percentage applications.

Before next week:

Try tonight's take-home practice. Want more? GED.com's free test preview tutorials (tools.ged.com) are always open, in English or Spanish.